

RITWIJ ARYAN PARMAR

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Available full-time immediately | MS completed May 2026

Software Engineer building backend and ML systems for LLM inference, GenAI evaluation, cloud services, and AI infrastructure.

TECHNICAL EXPERIENCE

Distributed Robotics and Networked Embedded Sensing (DRONES) Lab, Buffalo, NY

Jul 2025 – Feb 2026

Research Aide

- Built a computer-vision inspection system for infrastructure culvert assessment, converting field-style inspection videos into defect masks and reviewable polygon annotations across 1,400+ sampled frames using SAM-3 prompts for structural defects and water/sediment conditions.
- Measured approximately 0.74 mask IoU and 2.3 seconds p95 keyframe inference after deploying the segmentation service on a Hugging Face GPU endpoint with a custom PyTorch container, versioned artifacts, batch frame inputs, endpoint logs, and per-run IoU/F1 reports.

Tata Elxsi – Global automotive engineering firm, Bengaluru, India

Jan 2024 – Jun 2024

Software Engineer Intern

- Built an ADAS evaluation-data pipeline that converted simulator runs into detector-ready datasets, aligning RGB frames, depth maps, semantic labels, ego pose, and calibration metadata into validated 2D/3D annotation packages.
- Automated heuristic failure mining across 120+ night, rain, occlusion, cut-in, and pedestrian-crossing simulation runs by flagging low-confidence detections, missed ground-truth matches, and label/calibration mismatches, improving held-out simulator pedestrian/vehicle recall from 0.71 to 0.82.

LLMate.ai – Seed-stage GenAI analytics startup, Bengaluru, India

Apr 2023 – Jul 2023

Software Engineer Intern

- Built a natural-language SQL backend over 50,000+ customer, transaction, and activity records, enforcing schema retrieval, SQL parser validation, read-only execution, row/time limits, and query-cost controls before database execution.
- Built a query reliability layer with dry-run checks, database-error tracing, clause-level repair, duplicate-request suppression, result caching, and 180+ replayable eval questions, reducing failed runs from about 31 percent to 12 percent and repeated-request latency by about 40 percent.

PROJECTS

HelixServe – Low-Latency LLM Inference Server

GitHub | Demo

- Increased LLM serving throughput 5.7x from 175.8 to 1,007.8 tokens per second and reduced p95 latency 75 percent from 3.28 seconds to 0.80 seconds on GCP NVIDIA L4 by building a low-latency serving engine with paged KV cache, continuous batching, prefix caching, and CUDA Graph decode.

ManoVarta – Multilingual GenAI Evaluation Workflow

GitHub | Demo

- Built and deployed a multilingual GenAI assessment workflow on Google Cloud Run, combining dialogue state, evidence extraction, structured scoring, safety routing, and replay evaluation over 60 dual-annotated sessions and 48 archived conversations, reaching 78.3 percent coverage completeness, 63.9 percent exact match, and 0 parse failures.

SRE-Nidaan – Incident Response Backend with vLLM

GitHub | Demo

- Built and deployed a 3-microservice incident-response backend on Google Cloud Run using Next.js, FastAPI, and vLLM, integrating 9 APIs for telemetry analysis, runbook retrieval, remediation gating, analyst feedback, and LLM evaluation over 2,500 incident scenarios.

OPEN-SOURCE ENGINEERING

Open-Source Contributor – NVIDIA Dynamo, Firecrawl, Agno, Promptfoo, Roboflow Supervision, Qdrant, Unsloth

- Contributed 30+ merged upstream PRs to externally reviewed AI infrastructure projects across LLM inference, agent runtimes, eval platforms, vector search, computer-vision metrics, and model-serving tooling.
- Fixed cache-key correctness in NVIDIA Dynamo's distributed LLM inference stack by replacing duplicated Mooncake token-block hashing with the shared runtime KV-hashing contract, preventing data-generation traces from drifting from production cache semantics.
- Fixed production reliability and evaluation bugs across Firecrawl, Agno, Unsloth, Promptfoo, Roboflow Supervision, and Qdrant: Redis client leaks, MCP session cleanup, tool-output crashes, grader routing, non-square OBB IoU errors, and scalar-quantized L2 score shifts.

SKILLS

Languages: Python, C++, Java, SQL

ML & Inference: PyTorch, LLMs, RAG, Computer Vision, vLLM, CUDA, GPU Computing

Backend & Cloud: FastAPI, Spring Boot, Docker, GCP, Cloud Run, BigQuery, RabbitMQ, Distributed Systems, Observability, CI/CD

EDUCATION

University at Buffalo – SUNY

Jan 2025 – May 2026

Master of Science in Computer Science and Engineering, GPA: 3.74 out of 4.0

Manipal Institute of Technology

Aug 2020 – Aug 2024

Bachelor of Technology in Computer Science and Engineering